Example Institute of Technology

B.Tech Semester III Examination, November 2025

CS301 Data Structures

Time: 3 Hours Maximum Marks: 34

PART A

Answer all questions. Each question carries 2 marks.

1. Define a stack and list its operations.

2. What is a linked list?

3. What is a hash collision?

PART B

Answer any two questions.

4. (a) Explain how a queue can be implemented using a circular array, showing the enqueue and dequeue operations. [6]

(b) Convert the infix expression A+B*C into postfix using a stack. [4]

5. Explain insertion and deletion in a binary search tree with an example. [10]

6. Write the algorithm for breadth first search of a graph and explain its time complexity. [10]

PART C

7. Explain merge sort and analyse its time complexity. [8]